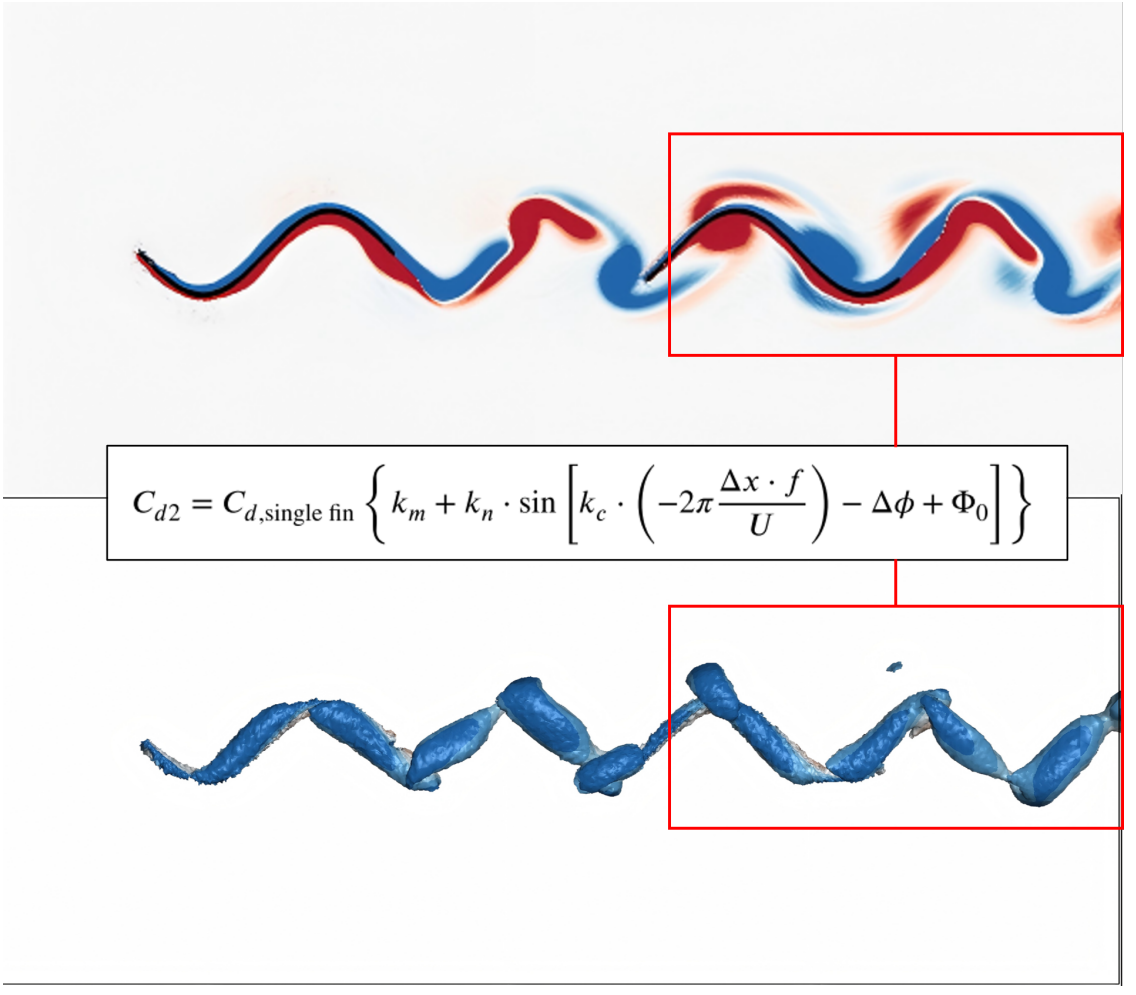
Graphical Abstract

Optimized design and investigation about propulsion of bionic Tandem undulating fins II : The coupled effect of phase difference and fin spacing

Wenzhuo Li, Ming Xu, Shunxiang Pang, Fenghua Qin, Jun Jing, Wan Cheng, Xisheng Luo



Highlights

Optimized design and investigation about propulsion of bionic Tandem undulating fins II : The coupled effect of phase difference and fin spacing

Wenzhuo Li, Ming Xu, Shunxiang Pang, Fenghua Qin, Jun Jing, Wan Cheng, Xisheng Luo

- Physical rationality of the two-dimensional model simplification.
- Two-dimensional flow field interference patterns and the proposal of the equivalent phase model.
- Validation of the semi-empirical equivalent phase model's applicability in three-dimensional flow fields.

Optimized design and investigation about propulsion of bionic Tandem undulating fins II : The coupled effect of phase difference and fin spacing

Wenzhuo Li[a], Ming Xu[a], Shunxiang Pang[a], Fenghua Qin[a,*], Jun Jing[b,**], Wan Cheng[a] and Xisheng Luo[a]

[a]Department of Modern Mechanics, University of Science and Technology of China, 230026, Hefei, China
[b]China Ship Development and Design Center, Wuhan, 430064, China

ARTICLE INFO

Keywords:
tandem undulating fins
propulsion performance
fin spacing
equivalent phase

ABSTRACT

Undulating fin propulsion offers advantages such as high maneuverability, strong stability, and good stealth, but its thrust output remains relatively low. In nature, schooling fish can effectively utilize wakes to significantly enhance their individual swimming speeds. Inspired by this, a similar beneficial fluid coupling effect may exist between undulating fins. In this paper, using the computational fluid dynamics software Palabos, two-dimensional and three-dimensional numerical simulations are carried out for tandem long fins employing purely undulatory propulsion, which intentionally restricts spanwise flow. Compared with the three-dimensional simulation results, the two-dimensional simulations effectively capture the primary characteristics of the wake vortex structures, while the significantly reduced computational cost makes them highly suitable for parametric studies. The effects of phase difference and fin spacing between the anterior and posterior fins on propulsion performance are systematically analyzed. The results demonstrate that while the tandem arrangement has no significant effect on the thrust of the anterior fin, the thrust of the posterior fin exhibits a distinct sinusoidal trend with variations in phase difference and fin spacing. Crucially, our analysis reveals that although the effects of spatial fin spacing and temporal phase difference on wake interference are physically decoupled, their combined impact can be uniformly described by introducing the concept of an "equivalent phase". Based on this concept, a mathematical model capable of predicting the thrust variation of the posterior fin is established. This model quantitatively elucidates the mechanism of wake vortex merging for undulating fins and effectively predicts the variation of tandem undulating fin thrust with phase difference and fin spacing, thereby laying a theoretical foundation for the optimal design of high-thrust undulating fins.

1. Introduction

To address complex engineering challenges, researchers frequently draw inspiration from biological models, systems, and elements. Applying biological principles to the design and manufacture of artificial machinery holds significant appeal for engineers. Based on their propulsive structures, the swimming modes of fish are primarily categorized into two types: median and/or paired fin (MPF) propulsion and body and/or caudal fin (BCF) propulsion (Salazar et al., 2018; Krishnadas et al., 2018; Sfakiotakis et al., 1999). In terms of propulsive speed, the BCF mode is capable of generating greater thrust and instantaneous acceleration. However, from the perspective of maneuvering stability, fish employing the MPF mode are better adapted to complex underwater environments.

Over the past few decades, numerous biomimetic studies have been dedicated to the optimization of engineering designs (MacIver et al., 2004). In the field of underwater biomimetic robotics, many successful engineering applications have emerged (Yuh, 2000). Autonomous underwater vehicles (AUVs) represent a quintessential example (Bandyopadhyay, 2005; Kruusmaa et al., 2014; Costa et al., 2018b). To date, researchers have designed and fabricated various prototypes, particularly those tailored for cruising missions. For instance, Mwaffo et al. (2017) investigated the flow field structure during zebrafish swimming through particle image velocimetry (PIV) experiments to elucidate the evolutionary advantages of burst-and-coast swimming. Chan et al. (2007) examined the effects of flapping frequency

*Corresponding author
qfh@ustc.edu.cn (F. Qin); 47531313@qq.com (J. Jing)
ORCID(s): 0009-0001-7608-2992 (W. Li); 0000-0003-1728-425X (F. Qin); 0000-0003-3960-4162 (W. Cheng); 0000-0002-4303-8290 (X. Luo)

and amplitude on the swimming speed of a boxfish-inspired robotic fish. Costa et al. (2017, 2018a,c) designed a boxfish-like robot and conducted corresponding numerical simulations, predicting its underwater motion via multibody dynamic techniques. Furthermore, they established experimental setups to investigate its propulsive performance, thereby validating the numerical predictions. Nevertheless, existing systems still struggle to meet the escalating demands for agility and hovering capabilities (Sfakiotakis et al., 2015). Moreover, in military applications such as anti-submarine warfare and reconnaissance, critical factors including wake stealth, silent propulsion, and maneuverability must be taken into account.

Compared to current artificial vehicles, certain fish species (e.g., the black ghost knifefish) exhibit remarkable advantages in efficiency and maneuverability owing to their elongated fins. Studies have revealed that their long dorsal and/or anal fins undulate in a sinusoidal wave-like manner from head to tail, while the body moves in the opposite direction (Hu et al., 2009). Blake (1983) established a kinematic model for forward-swimming fish, pioneering the study of undulatory fin propulsion. Subsequently, Lighthill and Blake applied slender-body theory to the hydrodynamic modeling of undulating fins (Lighthill and Blake, 1990; Lighthill, 1990a,b,c). Shirgaonkar et al. (2008) investigated the hydrodynamic characteristics of undulating fins and measured the forces generated, demonstrating that traveling waves on a single fin can simultaneously produce thrust and lateral force. Utilizing PIV, Neveln et al. (2014) measured the flow field around an undulating fin and developed a computational fluid dynamics (CFD) model to simulate its hydrodynamics; their results showed that a series of vortices are shed from the long edge of the fin as the undulatory wave propagates along it. Building upon the understanding of undulatory fin propulsion mechanisms, numerous researchers have designed prototypes driven by either a single biomimetic undulating fin (Epstein et al., 2006; Guo et al., 2003; Hu et al., 2009; Liu and Curet, 2018; Low, 2006) or bilaterally symmetric fins (Kim et al., 2017; Shang et al., 2012; Wang et al., 2017). Research has indicated that the magnitude of the lateral force increases significantly with rising thrust. This also reveals a primary limitation of undulating fins: to maintain high efficiency and maneuverability, the maximum thrust producible by a single fin is often constrained by fluid dynamics, meaning that thrust enhancement cannot be achieved merely by increasing parameters such as fin size.

Certain fish are capable of harvesting maximal energy from their surrounding environment to achieve long-distance cruising, high-speed, or highly efficient swimming. The interaction between these fish (or schools of fish) and ambient flows has attracted widespread research attention (Becker et al., 2015; Tian et al., 2016; Dai et al., 2018; Ramananarivo et al., 2016; Zhu et al., 2014). Liao et al. (2003) placed a live fish in the wake of a D-shaped cylinder, finding that the fish's energy expenditure in the bluff-body wake was lower than in free-stream conditions. Subsequently, Beal et al. (2006) conducted similar experiments using dead fish, revealing that the carcass was propelled upstream due to resonance with wake vortices. Pan et al. (2010) then investigated the hydrodynamics of a flexible plate in the wake of a D-shaped cylinder, observing that the plate initially underwent large-amplitude oscillations before developing a lateral inclination. Jia and Yin (2008) performed experiments in a flowing soap film to illustrate the interactions between two tandem flexible filaments, discovering that under the influence of vortex shedding, the downstream filament experienced greater hydrodynamic forces than either the upstream filament or a single filament in uniform flow. Additionally, experimental analyses of multi-fin systems have been conducted to verify inter-fin interactions (Arreola and Westneat, 1996; Hove et al., 2001; Lauder et al., 2007). These compelling studies suggest that hydrodynamic coupling between biomimetic undulating fins (the schooling effect) can be leveraged to enhance propulsive performance.

To fully exploit the beneficial hydrodynamic interactions between tandem undulating fins, recent studies have focused on the hydrodynamic performance of tandem undulating fin systems. Pang et al. (2021) conceptualized a prototype aquatic biomimetic vehicle propelled by four distributed undulating fins and systematically investigated the effect of phase difference on tandem propulsive force, finding that vortex interactions between fins at specific phase differences could significantly boost overall thrust. Wei et al. (2024) employed a high-resolution constrained immersed boundary method to simulate a three-dimensional (3D) dual-fin system, focusing on the impacts of array arrangement, phase difference, and spatial distance on hydrodynamic performance; they noted that tandem configurations substantially enhance aft-fin thrust—a phenomenon closely linked to leading-edge suction mechanisms—though the effect of phase difference was not pronounced. Similarly, Li et al. (2024) conducted an in-depth exploration of the hydrodynamic characteristics of tandem multi-wing systems. However, although these studies have identified the individual effects of phase difference or spacing, most existing research has examined these two parameters in isolation, lacking a comprehensive understanding of the synergistic mechanisms between phase difference and fin spacing.

Motivated by this gap, the present study focuses on the coupled effects of phase difference and fin spacing on the overall flow field and hydrodynamic performance in tandem undulating fin arrangements. We aim to uncover the fluid

mechanical mechanisms underlying tandem fin interactions and elucidate the evolutionary patterns of vortex merging or cancellation, thereby providing theoretical support and optimization strategies for further enhancing the propulsive performance of biomimetic undulating robots.

The geometric model, kinematic formulations, and computational setup employed in this study are presented in Section 2. Simulation results detailing the effects of fin spacing and phase difference on tandem undulating fins are provided in Section 3. Finally, conclusions are drawn in Section 4.

2. Problem formulation and methods

2.1. Kinematics and geometric model of undulating fins

Pang et al. (2021) demonstrated that the hydrodynamic coupling between anterior and posterior fins is primarily mediated through the merging or cancellation of their respective shed vortices in the wake. Therefore, the present work specifically investigates how variations in inter-fin spacing (Δx) and phase difference ($\Delta\phi$) influence the interaction—particularly the merging and cancellation—of these trailing vortices, as well as the resulting propulsive performance specifically thrust generation.

To this end, a 3D purely undulatory fin model is established as the primary numerical framework (Figure 1). It should be noted that, unlike the combined undulatory–oscillatory motion of the bionic inspiration, the black ghost knifefish, the present 3D model implements purely undulatory motion only. The absence of spanwise deflection in the wake generated by the purely undulatory fin facilitates a clearer visualization and analysis of the evolution and interaction of 3D vortex structures. *A comparative investigation into the differences in flow mechanisms between the purely undulatory fin and the combined undulatory–oscillatory fin will be addressed in future work.*

The 3D undulating fin model is constructed within a right-handed Cartesian coordinate system (xyz), with the fins aligned along the x-axis and uniformly extruded in the z-direction over a spanwise width w ($w = 0.05$ m). The leading edge of the anterior fin is positioned at $x = 0$, and the horizontal distance between the trailing edge of the anterior fin and the leading edge of the posterior fin is defined as the inter-fin spacing, Δx. Both fins undergo sinusoidal undulatory motion which is expressed as

$$\begin{aligned} y_1(x,z,t) &= A\sin\left(kx-\omega t+\phi_1\right), \quad 0\le x\le L, \quad -w/2\le z\le w/2,\\ y_2(x,z,t) &= A\sin\left[k(x-L-\Delta x)-\omega t+\phi_2\right], \quad 0\le x-L-\Delta x\le L, \quad -w/2\le z\le w/2 \end{aligned} \tag{1}$$

where subscripts 1 and 2 denote the anterior and posterior fins, respectively; A is the undulation amplitude ($A = 0.06$ m), $\omega = 2\pi/T$ is the angular frequency, $f = 1/T$ is the undulating frequency ($f = 2$ Hz), $k = 2\pi n/L$ is the wavenumber, ϕ is the initial phase, L is the initial length of each fin ($L = 0.362$ m), $n = L/\lambda$ is the wave number along the fin ($n = 1$), and T is the oscillation period. The uniform inflow velocity is set to $U = 0.5$ m/s (unless otherwise specified). In this study, identical values of A, ω, and k are prescribed for both fins. The initial phase of the anterior fin (ϕ_1) is fixed at zero, rendering the initial phase of the posterior fin (ϕ_2) equivalent to the phase difference $\Delta\phi = \phi_2 - \phi_1$. Parametric studies are conducted by systematically varying $\Delta\phi$ and Δx while holding all other kinematic and geometric parameters constant.

To conduct efficient parametric screening and validate the semi-empirical equivalent phase model, a two-dimensional (2D) simplified model is further derived as a cross-sectional slice of the 3D fin along the mid-span plane (xy-plane at $z = 0$). This 2D model inherits the identical cross-sectional geometry and kinematics from the 3D model, serving to verify that the 2D simplification adequately captures the fundamental mechanisms of wake vortex interactions without sacrificing physical accuracy.

2.2. Numerical methodology

In this study, numerical simulations for both 3D and 2D fin models are performed using the open-source Lattice Boltzmann Method (LBM) software library, Palabos Latt et al. (2021). To accurately resolve the unsteady turbulent flow and complex trailing vortex interactions, an explicit Large Eddy Simulation (LES) framework integrated with a regularized Bhatnagar–Gross–Krook (BGK) collision operator is employed. The spatial discrete velocity space is discretized using the standard D3Q19 lattice descriptor for the 3D domain and the D2Q9 descriptor for the 2D domain. The regularized scheme filters out non-equilibrium higher-order ghost moments to ensure numerical stability at high Reynolds numbers, while subgrid-scale (SGS) turbulence dissipation is modeled via the explicit Smagorinsky model with a Smagorinsky constant of $C_s = 0.12$. An Immersed Boundary Method (IBM) based on Inamuro's iterative velocity-correction algorithm is applied to capture the fluid–structure interaction of the moving fin surfaces.

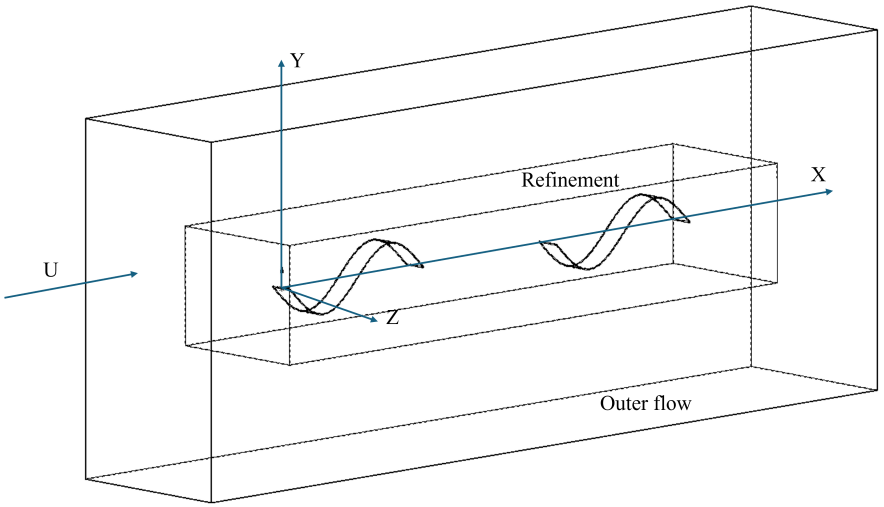


Figure 1: Computational domain for the tandem undulating fins model.

The 3D computational domain, as illustrated in Fig. 1, consists of two regions: the outer flow field and the localized refinement zone. Expressed in terms of the fin length L ($L = 0.362$ m), the outer flow field is a cuboid extending from $(-1.5L, -L, -L)$ to $(4.5L, L, L)$, while the refinement zone spans approximately from $(-0.8L, -0.4L, -0.8L)$ to $(2.8L + \Delta x, 0.4L, 0.8L)$.

The 2D computational domain is divided into three regions: the outer flow field, the wake refinement zone, and the body-fitted refinement zone. Expressed in terms of L, the outer flow field is a rectangular region extending approximately from $(-4L, -4L)$ to $(14L, 4L)$. The wake refinement zone spans from $(-2L, -2L)$ to $(12L, 2L)$, and the body-fitted refinement zone extends from $(-0.7L, -0.7L)$ to $(3.4L + \Delta x, 0.7L)$.

A tethered model is adopted for both 3D and 2D configurations, with a uniform incoming flow velocity U specified at the inlet. Dirichlet velocity boundary conditions are imposed at the inlet, while constant pressure conditions are defined at the outlet. Free-slip boundary conditions are applied on the four surrounding walls, while exact no-slip conditions on the moving fin surfaces are enforced via IBM. The fluid is assumed to be incompressible and Newtonian with a density of $\rho = 998.2$ kg/m^3 and a dynamic viscosity of $\mu = 1.01 \times 10^{-3}$ Pa $\cdot$ s. The characteristic Reynolds number is defined as:

$$Re = \frac{\rho U L}{\mu} \tag{2}$$

Depending on the prescribed inflow conditions across different cases, the Reynolds number in this study ranges from 10^5 to 10^6. To adhere to the weakly incompressible limit in LBM ($Ma \ll 1$), the physical time step Δt is controlled such that the maximum characteristic lattice velocity satisfies $u_{\mathrm{LB,max}} \leq 0.08$. Hydrodynamic propulsive forces acting on the fin surfaces are evaluated directly at each time step using multi-iteration Lagrangian IBM force integration.

2.3. Grid independence

To verify the grid independence of the 2D simulations, three sets of meshes (coarse, medium, and fine) were generated, as listed in Table 1. The parameters for the verification case were set as follows: $f = 2$ Hz, $n_1 = n_2 = 1$, $A = 0.06$ m, $L = 0.362$ m, $U = 0.5$ m/s, a phase difference of zero, and an inter-fin spacing $\Delta x = L$. Since the propulsive thrust (defined as the streamwise component of the hydrodynamic force acting on the fin surfaces, denoted as F_x) is significantly smaller than the lateral force and this study primarily focuses on propulsive performance, the time history of F_x was used to evaluate the mesh quality. As shown in Fig. 2(a), the F_x curves obtained from the medium and fine meshes are relatively smooth with minor discrepancies between them, whereas the coarse mesh

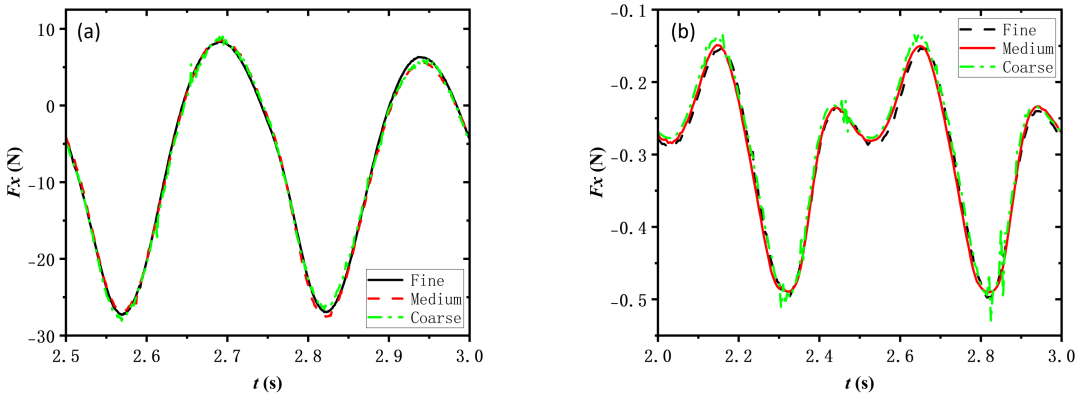


Figure 2: Time histories of the propulsive thrust under coarse, medium, and fine mesh resolutions for grid independence verification: (a) 2D model and (b) 3D model.

exhibits considerable fluctuations and deviates substantially from the fine-mesh results. The time-averaged F_x values computed by the fine, medium, and coarse meshes are 7.81 N, 7.86 N, and 7.95 N, respectively. Considering both accuracy and computational cost, the medium mesh was selected for the subsequent 2D simulations.

Table 1
Mesh configurations and grid sizes for the spatial independence verification.

Type	Mesh size	2D Cells	3D Cells
Coarse	2 mm	182,995	746,229
Medium	1 mm	714,171	3,553,920
Fine	0.5 mm	859,400	7,837,925

For the 3D cases, grid independence was verified using the three mesh resolutions detailed in Table 1. To ensure consistency, the validation cases were primarily based on configurations from prior studies. For the grid independence check and numerical method validation, the baseline parameters were set as follows: undulation frequency $f = 2$ Hz, wavenumber $n = 1$, undulation amplitude $A = 0.06$ m, fin length $L = 0.362$ m, and spanwise width $w = 0.05$ m. Notably, the inflow velocity U varied depending on the specific validation scenario: it was set to 0.4 m/s for the fin spacing validation and 0.3 m/s for the frequency variation validation, strictly aligning with the reference configurations. Furthermore, for our independent parametric investigation into the effect of phase difference (Case 2), the inflow velocity was uniformly set to 0.5 m/s. Similarly, the time history of F_x was employed to assess mesh adequacy. As illustrated in Fig. 2(b), the F_x histories from Fine and Medium are smooth and show small differences, while the result from Coarse displays significant oscillations near the peak values. The mean F_x values predicted by Fine, Medium, and Coarse are 0.288 N, 0.292 N, and 0.300 N, respectively. The relative error in the mean F_x between Medium and Fine is only 1.4%. Taking into account both numerical accuracy and computational efficiency, the Medium mesh was adopted for the 3D undulating fin simulations.

To validate the accuracy of the present numerical model and method, a simulation reproducing the experimental case of Zhou et al. (2010) was conducted for comparison. The specific parameters were: fin chord length $L = 0.5$ m, number of waves along the fin $n = 1.0$, undulation frequency $f = 2.0$ Hz, undulation amplitude $A = 0.02$ m, and uniform inflow velocity U varying from -2 to 2 m/s. Figure 3(a) compares the computed time-averaged thrust ($\overline{F}_x$) against the data reported by Zhou et al. (2010). It can be observed that the thrust values at various inflow velocities agree well with the reference results, demonstrating that the numerical model and methods employed in this study are reliable and appropriate.

To validate the accuracy and reliability of the present 3D numerical solver, a benchmark simulation of a single 3D undulating fin was conducted and compared with the experimental and numerical results reported by Zhang et al.

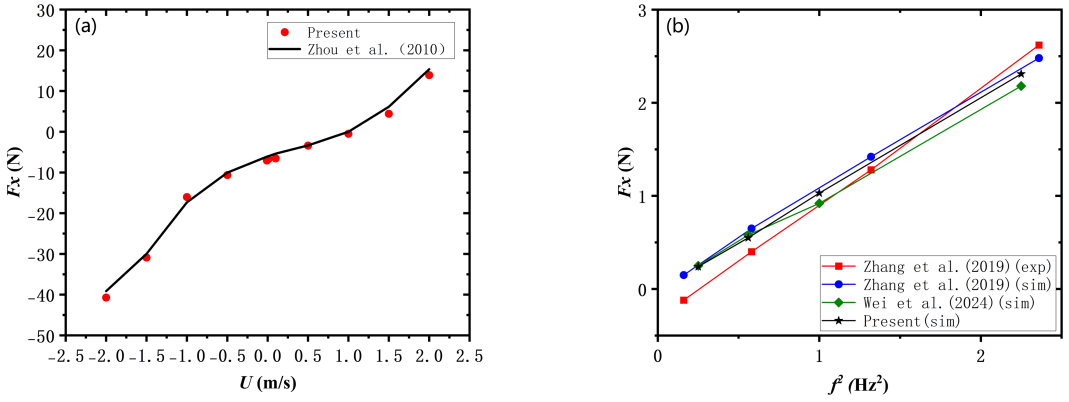


Figure 3: Validation of the numerical solver accuracy against existing literature: (a) 2D thrust at varying inflow velocities, and (b) 3D period-averaged thrust versus squared undulating frequency (f^2).

(2019) and Wei et al. (2024). The geometric and kinematic parameters of the 3D fin were set identical to those in the reference literature: fin length $L = 800$ mm, fin height $h = 50$ mm, wavenumber $n = L/\lambda = 2.0$, and maximum angular deflection $\phi_m = 85°$. Simulations were performed across various undulation frequencies ($f = 0.5$, 0.75, 1.0, and 1.5 Hz). Figure 3(b) compares the calculated period-averaged net thrust ($\overline{F}_x$) as a function of the square of frequency (f^2) against the experimental data from Zhang et al. (2019) and the numerical predictions from Wei et al. (2024). It can be observed that the thrust values across all frequencies agree well with both experimental and numerical literature data, effectively reproducing the characteristic linear relationship between average thrust and f^2. This excellent agreement demonstrates that the 3D numerical model, turbulence framework, and fluid–structure interaction method employed in this study are highly reliable and accurate for investigating the hydrodynamics of undulating fins.

3. Results

3.1. Hydrodynamic equivalence and justification of the 2D simplified model

First, the spanwise-averaged thrust obtained from the 2D undulating fin model is compared with that of the 3D model to validate the feasibility of the 2D simplification for a single fin. The parameters for the 2D case are set as follows: $A = 0.06$ m, $L = 0.362$ m, $n = 1$, $f = 2$ Hz, and $U = 0.5$ m/s. The 3D fin shares identical parameters, with a fin width of $w = 0.05$ m. The average thrust ($\overline{F}_x$) over a spanwise width of 0.05 m computed by the simplified 2D model is 0.256711 N, whereas the corresponding value from the 3D model is 0.292493 N. The relative error between the two models is approximately 12%, which is considered acceptably small, indicating that the 2D simplified model successfully captures the majority of the thrust generation. Figure 4 presents the vorticity fields obtained from both the 2D and 3D simulations. It can be observed that the 2D model effectively captures the wake vortices shed by the undulating fin but inherently fails to resolve the lateral vortices induced by 3D effects. In the 3D model, vortex shedding from both side edges (LEV) contributes additional thrust, which explains why the 2D model yields a slightly lower thrust prediction. Nevertheless, since the wake vortices account for the dominant portion of the total thrust, it is concluded that the 2D simplified model is sufficiently accurate for investigating the hydrodynamic interactions among tandem fins.

The numerical methods and grids described above are employed to conduct a comprehensive 2D parametric study and 3D validation. The parameter settings for the 2D undulating fin numerical simulations are presented in Table 2. Case 1 is used to investigate the effect of fin spacing on thrust, providing the foundation for deriving an empirical formula. Case 2 is designed to evaluate the isolated influence of phase difference; Case 3 evaluates the effect of wavenumber n; Case 4 assesses the impact of undulation amplitude A; and Case 5 examines the influence of undulation frequency f. The 3D cases serve to verify both the validity of the 2D simplification and the cross-dimensional generalizability of the 2D theory.

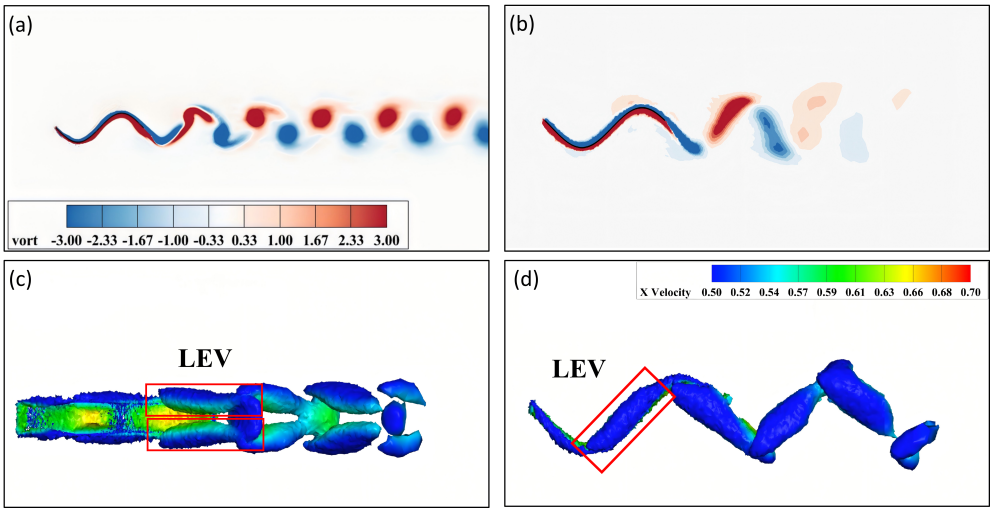


Figure 4: Comparison of instantaneous wake vortex structures between the 2D and 3D purely undulating fin models: (a) 2D vorticity field; (b) 3D mid-span vorticity field; and 3D vortex topology identified by the Q-criterion from (c) top view and (d) side view.

Table 2
Kinematic and geometric parameter configurations for the 2D parametric study.

Parameter	Case 1	Case 2	Case 3	Case 4	Case 5
U (m/s)	0.5	0.5	0.5	0.5	0.3
L (m)	0.362	0.362	0.724	0.362	0.362
n	1	1	2	1	1
f (Hz)	2	2	2	2	1
A (m)	0.06	0.06	0.06	0.12	0.06
$\Delta\phi$ (°)	-140	-180,-90,0,90,140	-140	-140	-140
$\Delta x/L$	[0.1,1.0]	[0.1,1.0]	[0.1,1.0]	[0.1,1.0]	[0.1,1.0]

3.2. 2D spatiotemporal interference mechanisms and the equivalent phase model

This study primarily focuses on the effects of fin spacing and phase difference on the thrust of undulating fins. To isolate these effects from other parameters, the dimensionless thrust coefficient of the undulating fin is defined as:

$$C_D = \frac{2F_x}{\rho U^2 A} \tag{3}$$

The comparison between the thrust coefficients of the anterior and posterior fins in Case 1 and that of a single fin is illustrated in Fig. 5. The black line with pentagram markers represents the thrust coefficient of the single fin; the blue squares denote the thrust coefficient of the anterior fin; and the red solid circles indicate the thrust coefficient of the posterior fin. It can be observed that when the ratio of fin spacing to fin length ($\Delta x/L$) varies from 0.1 to 1.0, the thrust coefficient of the anterior fin fluctuates only slightly around that of the single fin, indicating that the tandem arrangement has a negligible influence on the propulsive performance of the anterior fin. In contrast, the thrust coefficient of the posterior fin is generally lower than that of the single fin and exhibits a distinct sinusoidal variation trend with respect to $\Delta x/L$.

Figure 6 presents the vorticity contours and velocity magnitude distributions for different fin spacings in Case 1. At $\Delta x/L = 0.3$, the posterior fin passes directly through the center of the wake vortices shed by the anterior fin, leading to destructive interference between the vortices shed from the two fins and consequently reducing propulsive performance. At $\Delta x/L = 0.6$, the posterior fin again traverses the wake of the anterior fin, but in this configuration, the vortices shed from both fins merge constructively, thereby enhancing propulsion. For other values of $\Delta x/L$, the interaction between the anterior and posterior fin wakes exists in a transitional state where constructive superposition

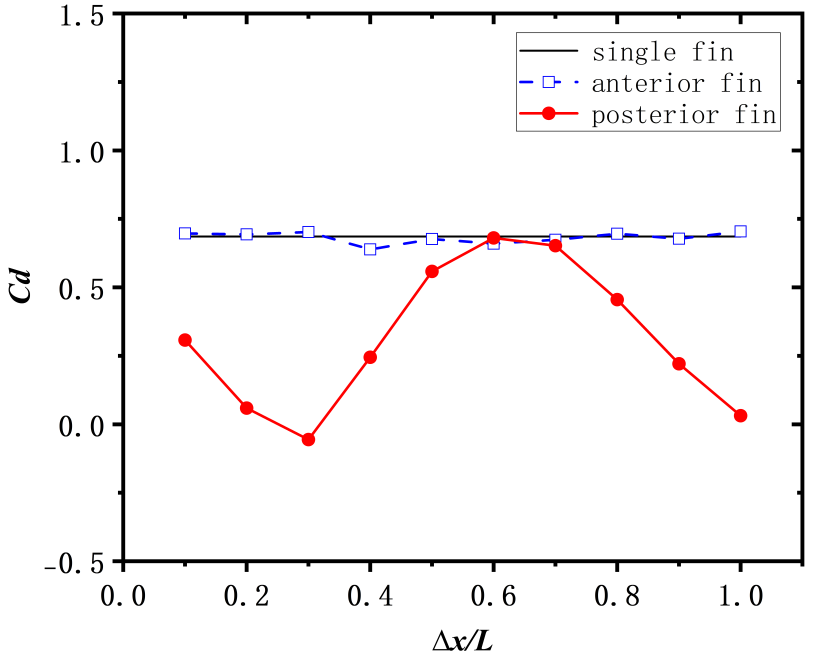


Figure 5: Variation of the averaged thrust coefficients for the anterior and posterior fins as a function of the inter-fin spacing ratio ($\Delta x/L$).

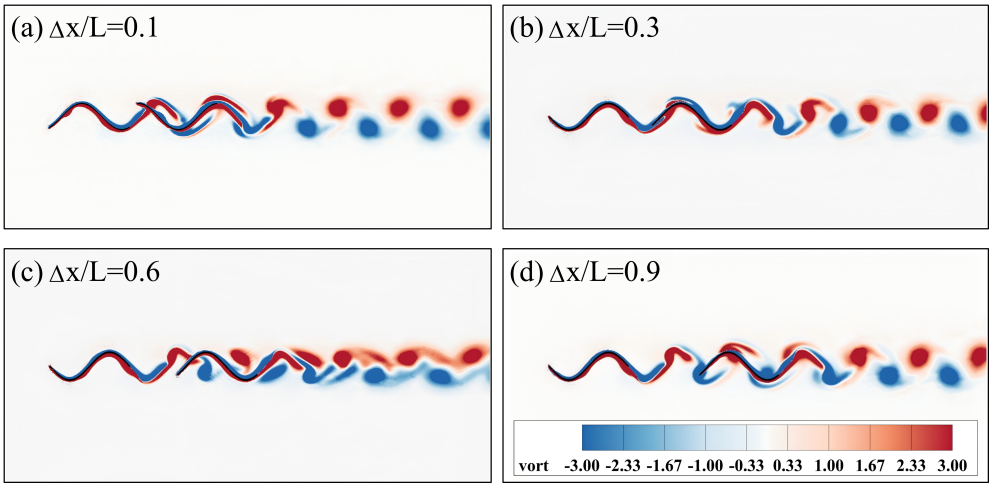


Figure 6: Instantaneous vorticity contours illustrating the alternating constructive and destructive wake interference patterns at different inter-fin spacing ratios ($\Delta x/L$).

and destructive cancellation coexist. From a fluid mechanics perspective, this sinusoidal variation of the posterior fin thrust is fundamentally governed by the relative positioning of the posterior fin within the spatially evolving vortex street shed by the anterior fin.

Combining this observation with the conclusions drawn by Pang et al. (2021), who reported that the posterior fin thrust also varies sinusoidally with phase difference, an empirical formula describing the variation of posterior fin

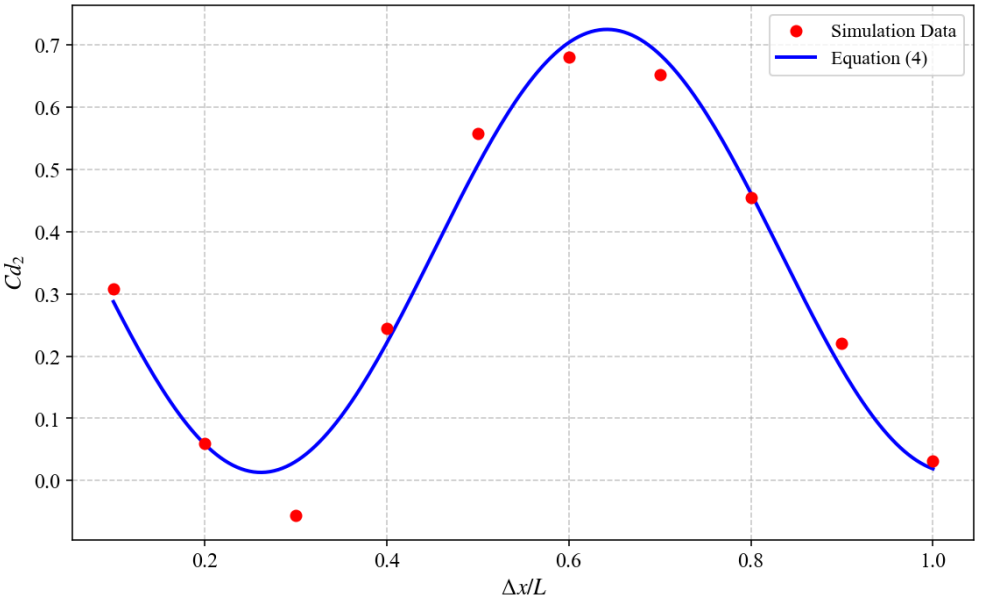


Figure 7: Comparison between the CFD-calculated posterior fin thrust coefficients and the analytical predictions of the equivalent phase model under varying spatial spacings.

thrust with both fin spacing and phase difference is derived through dimensional analysis:

$$C_{d2} = C_{d,\text{single fin}} \left\{ k_m + k_n \cdot \sin \left[k_c \cdot \left(-2\pi \frac{\Delta x \cdot f}{U} \right) - \Delta\phi + \Phi_0 \right] \right\} \tag{4}$$

where f is the undulation frequency, U is the inflow velocity, $\Delta\phi$ is the phase difference, Δx is the fin spacing, and k_m, k_n, k_c, and Φ_0 are undetermined coefficients.

The semi-empirical model given by Eq. 4 essentially serves as a macroscopic mathematical abstraction of the complex hydrodynamic processes involved in tandem undulating fin propulsion. The significant periodic fluctuation of the posterior fin thrust with respect to fin spacing and phase difference fundamentally stems from the unsteady interference between the wake vortices shed by the anterior fin and the kinematics of the posterior fin. Fin spacing and phase difference are the two critical parameters governing this interference. Accordingly, an equivalent phase is defined as:

$$\Phi_{eq} = k_c \cdot \left(-2\pi \frac{\Delta x \cdot f}{U} \right) - \Delta\phi + \Phi_0 \tag{5}$$

This represents a unified interference phase that incorporates the effects of spatial spacing and temporal phase, establishing a unified space–time scale and enabling the quantitative prediction of complex cascading hydrodynamic evolution patterns. As shown in Fig. 7, by fitting the four undetermined parameters to the simulation data from Case 1, the empirical formula achieves exceptional agreement with the CFD results.

The undulatory parameters not explicitly parameterized in the baseline empirical formula include the undulation amplitude A, while those yet to be validated encompass the phase difference $\Delta\phi$, wavenumber n, and undulation frequency f. To verify the robust applicability of the proposed empirical formula, Cases 2–5 were designed by varying these parameters.

The results of Case 2, which varies the phase difference between the anterior and posterior fins while keeping other parameters constant, are first presented as individual line fits in Fig. 8. It can be observed that across all six phase-difference configurations, the fitting curves agree remarkably well with the numerical solutions. As the phase difference varies, the thrust coefficient curve undergoes a highly regular rigid translation in the horizontal direction, whereas its baseline mean value and fluctuation amplitude remain largely unchanged. This intuitively confirms from a fluid

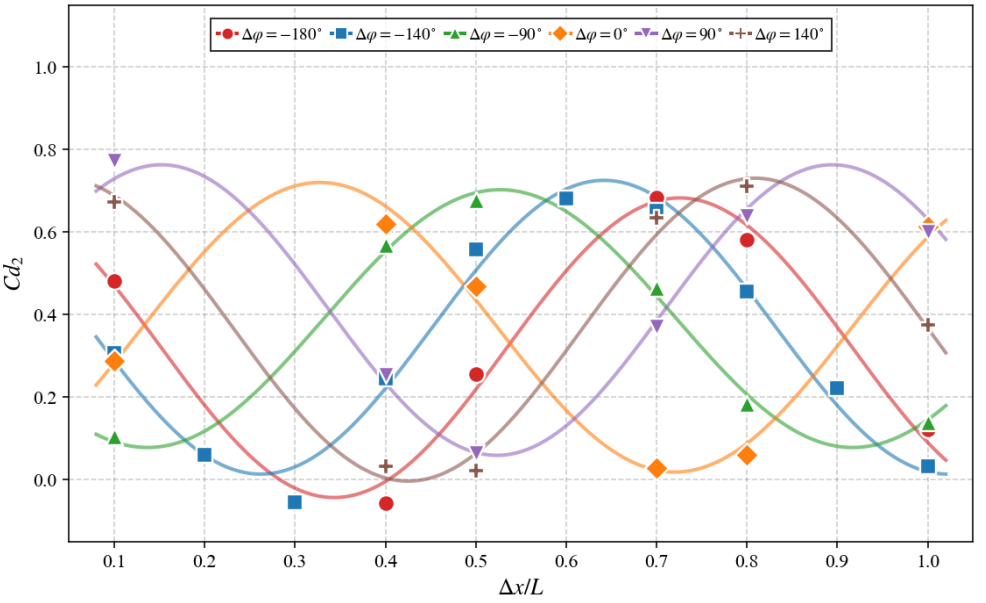


Figure 8: Effect of initial phase difference ($\Delta\phi$) on the posterior fin thrust coefficient, demonstrating the strict horizontal translation of the spatial interference pattern.

kinematics perspective the physical correctness of introducing the linear absolute term $-\Delta\phi$ in the formula—namely, that a temporal phase delay can be strictly mapped as an equivalent spatial phase shift of wake vortex interference.

To further validate the robustness of this mechanism and transition from 1D slices to a continuous parameter domain, Case 2 was systematically re-evaluated via simultaneous variation of the inter-fin spacing ratio ($\Delta x/L$) and the phase difference ($\Delta\phi$). As depicted in Fig. 9, a global 3D spatiotemporal coupling surface is constructed by jointly fitting all numerical simulation data points. The model surface matches the CFD data remarkably well ($R^2 = 0.971$), faithfully capturing the periodic wave-like distribution of the posterior fin's thrust coefficient.

Crucially, Table 3 compares the undetermined coefficients obtained from this global joint fit against those derived from the individual line fits. It is observed that not only do the phase-related parameters remain tightly bounded—with k_c fluctuating narrowly around $0.880 \sim 0.930$ (joint value: 0.909) and Φ_0 within $-1.848 \sim -2.091$ rad (joint value: -1.895 rad)—but the amplitude-related parameters k_m and k_n also exhibit remarkable stability, remaining closely anchored to their joint values of 0.543 and 0.509, respectively. This comprehensive parameter stability yields a profound physical insight: altering the inter-fin spacing and phase difference solely shifts the temporal and spatial alignment of the wake interference, without modifying the fundamental momentum injected into the wake or the absolute intensity of the vortex interactions. Consequently, the strong convergence between the global joint surface fit and individual line fits theoretically guarantees that the influences of spatial fin spacing and temporal phase difference on wake interference are physically completely decoupled, yet mathematically unified by the equivalent phase. Furthermore, the invariance of k_m and k_n in this context directly motivates and justifies the subsequent design of Cases 3 to 5; to rigorously test the generalizability of the proposed framework, it becomes imperative to deliberately alter the actual undulatory parameters (A, n, f, U) that actively dictate the wake momentum and interference amplitude.

To comprehensively assess the generalizability of the equivalent phase model under different hydrodynamic conditions, Cases 3 to 5 were conducted by varying the remaining undulatory parameters: wavenumber n (Case 3), undulation amplitude A (Case 4), and undulation frequency f (Case 5). The numerical simulation results and their corresponding empirical fits are consolidated in Fig. 10. As illustrated, whether the wake topology is altered by increasing the wavenumber, or the nonlinear flow separation and turbulent dissipation are intensified by a large undulation amplitude, or the propulsion system is pushed toward a critical weak-thrust regime by lowering the frequency, the empirical curves maintain exceptional agreement with the CFD scatter points. This demonstrates the robust adaptability of the theoretical framework across diverse oscillation intensities and kinematic configurations.

Table 3
Comparison of the undetermined coefficients derived from the global joint surface fit versus individual curve fits under varying temporal phase differences.

$\Delta\phi$ (°)	Fit method	R^2	k_m	k_n	k_c	Φ_0
-	Joint	0.971	0.543	0.509	**0.909**	**-1.895**
-180	Individual	0.984	0.466	0.530	**0.903**	**-1.892**
-140	Individual	0.976	0.538	0.519	**0.909**	**-1.848**
-90	Individual	0.994	0.569	0.456	**0.885**	**-2.041**
0	Individual	0.987	0.538	0.512	**0.880**	**-2.091**
90	Individual	0.999	0.599	0.514	**0.930**	**-1.854**
140	Individual	0.991	0.530	0.535	**0.899**	**-1.935**

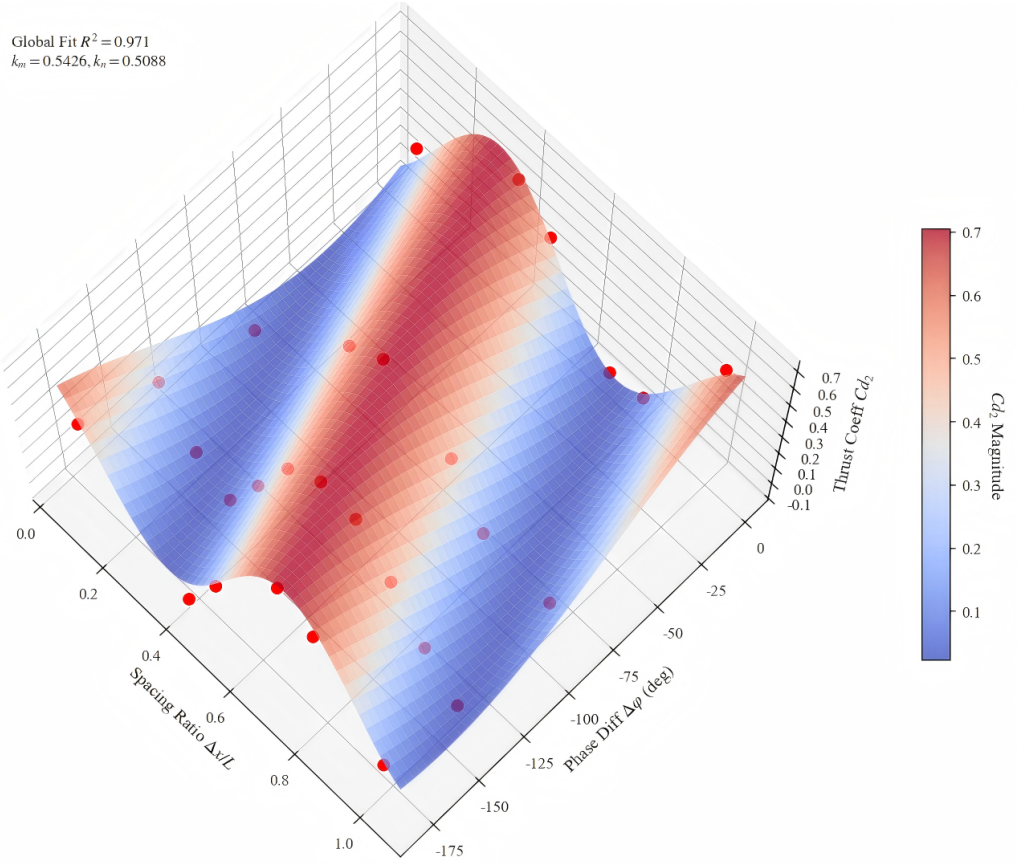


Figure 9: Global 3D spatiotemporal coupling surface mapping the posterior fin thrust coefficient as a simultaneous function of inter-fin spacing ratio and phase difference.

Table 4 summarizes the specific values of the four undetermined coefficients obtained from the fitting process across all test cases. A rigorous cross-comparison of these data reveals three core physical insights regarding the proposed model:

1. **Sensitivity of k_m and k_n:** The coefficients k_m and k_n exhibit noticeable variations when changing the wave parameters (A, n, f, U). This mathematically reflects that changes in fin kinematics directly govern the magnitude of momentum injected into the wake and the intensity of the shed vortices.
2. **Kinematic Invariance of k_c and Φ_0:** Conversely, the spatial convection correction coefficient k_c and the initial phase offset Φ_0 remain fundamentally stable and are practically unaffected by the alterations in undulatory

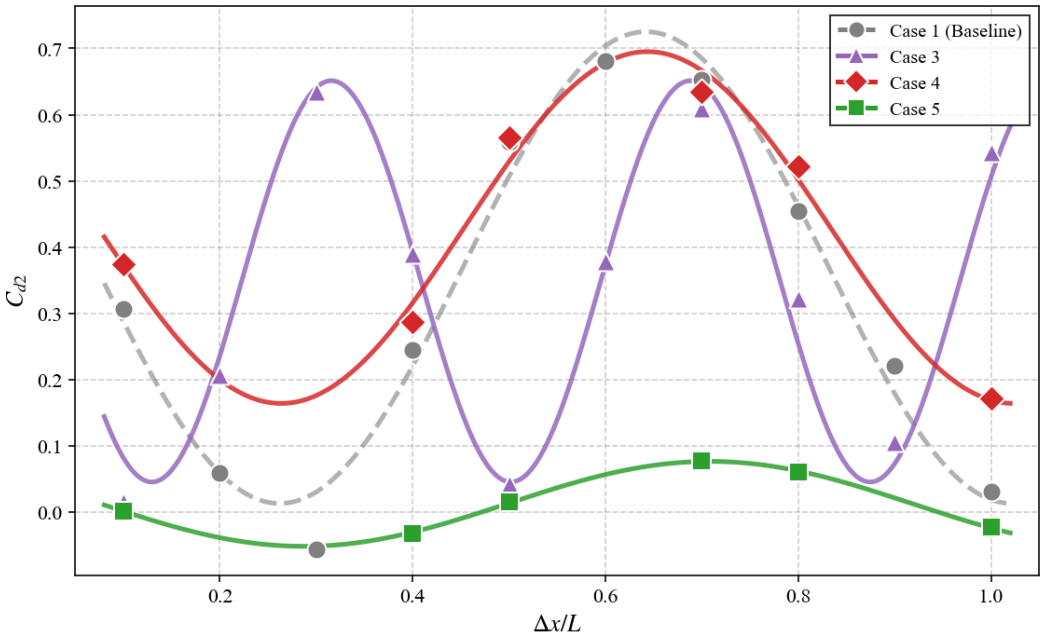


Figure 10: Validation of the equivalent phase model's robustness against variations in underlying undulatory parameters: wavenumber (Case 3), undulation amplitude (Case 4), and undulation frequency coupled with inflow velocity (Case 5).

parameters (e.g., k_c remains strictly around $0.9 \sim 0.98$, and Φ_0 stays within $-1.8 \sim -1.9$ rad). This remarkable stability proves that the equivalent phase (Φ_{eq}) is a robust kinematic invariant of the flow field. Once established, it reliably and universally characterizes the spatial and temporal interference mechanisms, irrespective of the underlying momentum injection variations.

3. **Theoretical Basis for 3D Validation:** Case 2 explicitly verified that the influences of phase difference and fin spacing are decoupled mechanisms. Consequently, we have a rigorous theoretical justification to extend this model into more complex 3D environments. By independently varying the spatial fin spacing and the temporal phase difference in the 3D flow field, we can individually validate whether the spatial and temporal interference components of the equivalent phase remain strictly applicable under strong spanwise spillage conditions.

Table 4
Summary of the fitted undetermined coefficients across all 2D parametric cases, highlighting the parameter sensitivity of k_m, k_n and the kinematic invariance of k_c, Φ_0.

Case number	R^2	k_m	k_n	k_c	Φ_0
Case 1	0.976	0.538	0.519	**0.909**	**-1.848**
Case 2	0.984	0.540	0.511	**0.909**	**-1.895**
Case 3	0.971	0.502	0.436	**0.927**	**-1.834**
Case 4	0.977	0.392	0.242	**0.909**	**-1.835**
Case 5	0.998	0.410	2.086	**0.984**	**-1.873**

3.3. Cross-dimensional validation of the equivalent phase model in 3D flow fields

Given the previously established decoupling between the spatial and temporal phases within the equivalent phase, the validity of the semi-empirical formula in 3D scenarios can be verified for each component separately. To maintain theoretical consistency with the 2D analysis and to reasonably isolate the geometric scaling effects in the 3D spatial

domain, a three-dimensional dimensionless thrust coefficient is defined as:

$$C_D = \frac{2F_x}{\rho U^2 A w} \tag{6}$$

where w represents the spanwise width of the 3D fin.

Fig. 11 presents a comparison between the posterior fin thrust coefficient obtained from numerical simulations and the prediction curve of the semi-empirical formula. Since the undulation direction is set along the positive x-axis, a negative coefficient corresponds to thrust, with its propulsive magnitude determined by the absolute value. The red discrete points denote the posterior fin thrust coefficients calculated from numerical simulations, while the black curve represents the semi-empirical formula fitted to these data points. It can be observed that when $\Delta x/L$ ranges from 0.3 to 0.8, the numerical simulation results agree remarkably well with the predictions of the semi-empirical formula.

Fig. 12 shows the velocity field contour around the 3D undulating fin at $\Delta x/L = 0.1$. At this close spacing, the inflow velocity experienced by the posterior fin is accelerated by the wake jet of the anterior fin, which directly compromises the thrust generation. This corresponds to the region in Fig. 11 where $\Delta x/L = 0.1$; the enhanced inhibitory effect of the anterior fin on the posterior fin at small spacings causes the actual thrust coefficient to fall below the theoretically predicted absolute value. Conversely, Fig. 13 illustrates the vorticity field around the 3D fin at $\Delta x/L = 1.0$. By the time the vortex rings shed from the anterior fin tail reach the posterior fin, they have largely dissipated. Consequently, the influence of the anterior fin wake vortices weakens, reducing the suppression on the posterior fin thrust and resulting in a thrust coefficient of greater magnitude than the predicted value. Notably, at $\Delta x/L = 0.5$, both the numerical simulation and the semi-empirical formula yield a posterior fin thrust coefficient whose magnitude explicitly exceeds the baseline coefficient of a single isolated 3D undulating fin ($C_D \approx 0.78$, corresponding to the absolute thrust of 0.29 N) under identical parameters. This firmly confirms that the coupling mechanism between the anterior fin and posterior fin wake vortices, as captured and described by our semi-empirical mathematical formula, can indeed enhance the overall propulsive performance in 3D environments. In summary, although minor discrepancies exist between the semi-empirical predictions and numerical simulations at extreme (very small or large) spacings, these deviations are physically explainable through wake jet acceleration and 3D vortex ring dissipation mechanisms. Within the intermediate spacing range, the semi-empirical formula shows excellent agreement with numerical results, thereby validating the cross-dimensional applicability of the spatial phase term in the 3D regime.

Subsequently, to investigate the applicability of the temporal phase term in 3D, the phase difference between the anterior and posterior fins was systematically varied while keeping all other kinematic and geometric parameters constant. Fig. 14 compares the numerical simulation results with the curve fitted by the semi-empirical formula. The thrust coefficients calculated under various phase differences align closely with the semi-empirical predictions. Moreover, at a phase difference of 240°, the posterior fin thrust coefficient once again surpasses the baseline value of the single isolated 3D fin, further corroborating that the direct hydrodynamic coupling between the anterior and posterior fins can significantly enhance posterior fin thrust. This specific validation case confirms that the linear term with a phase-difference coefficient of -1 in the semi-empirical formula seamlessly maintains its physical validity in 3D scenarios.

In summary, the series of validation results from the 3D undulating fin cases conclusively demonstrate the robust applicability of the proposed semi-empirical analytical model, constructed based on the "equivalent phase," in fully three-dimensional scenarios. From a physical mechanism perspective, the evolution of 3D tandem wakes is significantly more complex than that of 2D flow fields, heavily accompanied by strong spanwise spillage and the sophisticated 3D vortex ring roll-up and dissipation. Nevertheless, the mathematical structure established by this model still exhibits high self-consistency within the 3D flow field, successfully capturing the physical essence of unsteady wake interference.

Through its physically decoupled space–time architecture, the equation precisely maps the convection nature of wake vortices. In the spatial dimension, the convection wavelength term ($\frac{\Delta x \cdot f}{U}$) embedded within the formula confirms that, despite severe 3D diffusion, the energy core of the anterior fin wake vortices continues to propagate downstream in the form of stable traveling waves. In the temporal dimension, the phase difference enters the interference term with an absolute linear coefficient of -1 ($-\Delta\phi$), demonstrating that although 3D fluids exhibit viscous hysteresis, their dynamic response to initial phase differences strictly adheres to a rigorous same-frequency rigid translation rule.

More importantly, at intermediate spacings ($\Delta x/L = 0.5$) and specific phase matches ($\Delta\phi = 240°$), the predictions of the formula highly overlap with the CFD results, accurately pinpointing the interference extreme points that surpass the propulsive coefficient of a single isolated fin. Mechanistically, this phenomenon indicates that through equivalent

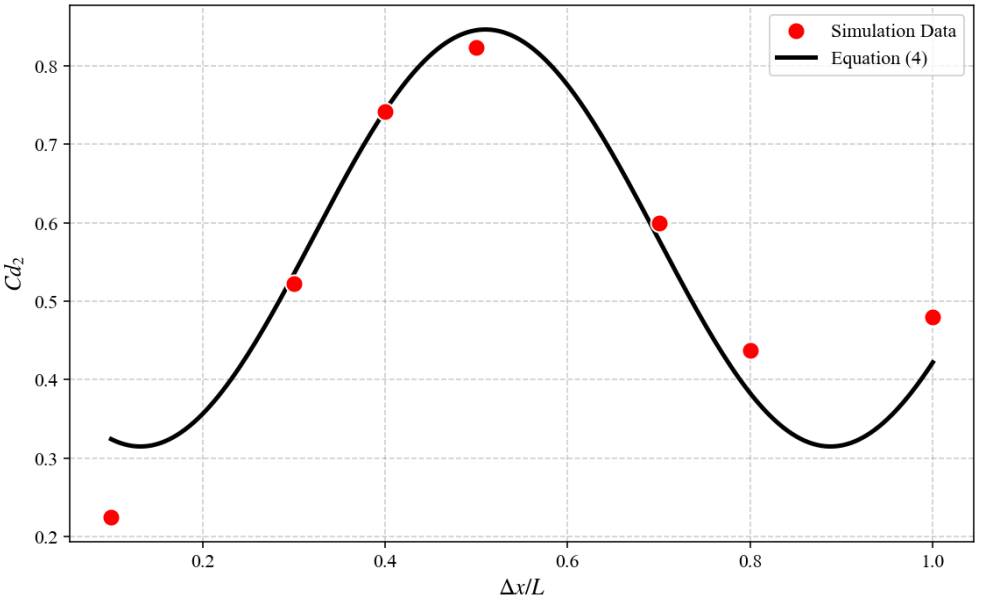


Figure 11: Comparison between the 3D CFD-calculated posterior fin thrust coefficients and the analytical equivalent phase model predictions under varying spatial fin spacings.

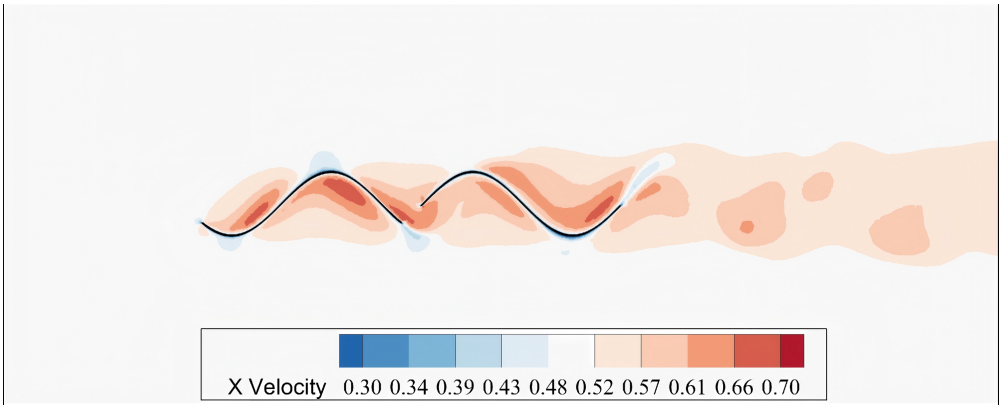


Figure 12: Instantaneous velocity magnitude contour revealing wake jet acceleration and posterior fin thrust suppression at a close spacing of $\Delta x/L = 0.1$.

phase matching guided by the formula, the posterior fin can perfectly avoid destructive vortex-core collisions and enter the highly unsteady flow field induced by the 3D vortex rings at an optimal attitude for energy harvesting. This fully proves that the semi-empirical equation successfully captures the underlying physical mechanisms of tandem biomimetic propulsion, providing a reliable theoretical framework for the space–time coupling optimization of future multi-body cooperative propulsion systems.

4. Conclusion

This study systematically investigates the hydrodynamic coupling and interference mechanisms of tandem undulating fins under a pure undulation mode using CFD, and proposes a semi-empirical analytical model to predict the thrust evolution of the posterior fin. The main conclusions are as follows:

1. **Physical rationality of the 2D model simplification.** By comparing the hydrodynamic characteristics of 2D and 3D pure undulating fins, the rationality and efficiency of the 2D model simplification in exploring longitudinal

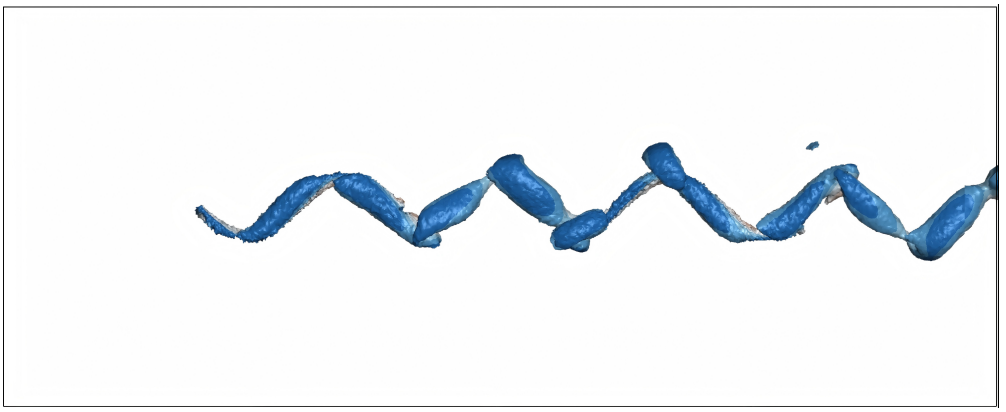


Figure 13: Instantaneous 3D vortex topology indicating vortex ring dissipation and weakened interference at a large spacing of $\Delta x/L = 1.0$.

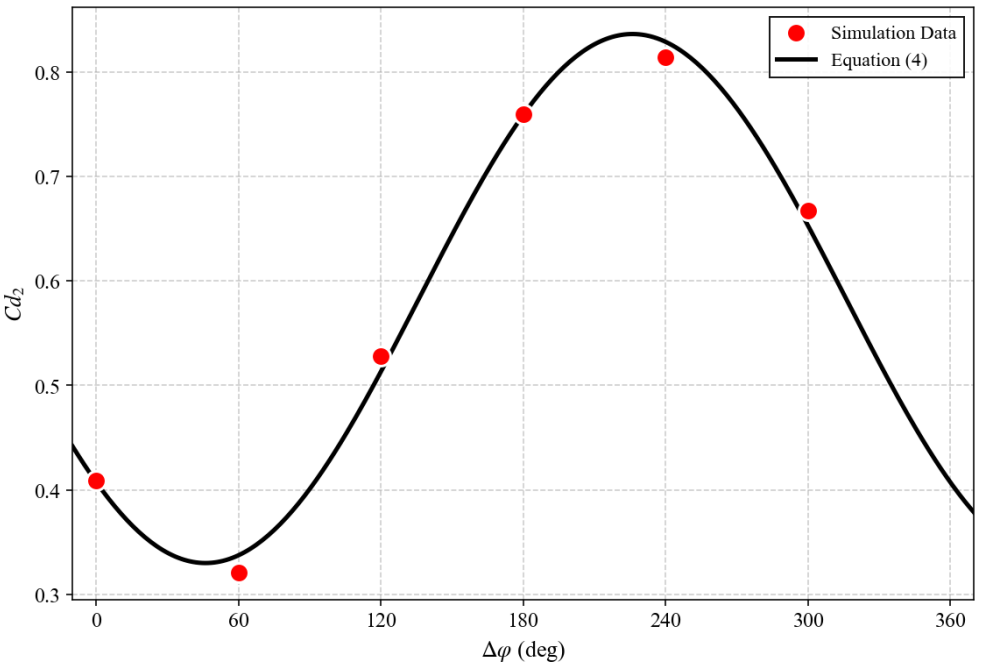


Figure 14: Comparison between the 3D CFD-calculated posterior fin thrust coefficients and the analytical equivalent phase model predictions under varying temporal phase differences at a fixed intermediate spacing ($\Delta x/L = 0.5$).

tandem interference mechanisms are verified. The results indicate that although the 2D model neglects 3D spillage and spanwise vortex shedding effects, leading to a thrust prediction deviation of approximately 12%, it accurately captures the core vortex shedding and longitudinal convection evolution characteristics within the anterior fin wake. The 2D model is sufficient to reflect the dominant mechanism of wake vortex interference and significantly reduces computational costs, thereby providing a reliable numerical foundation for conducting extensive parametric studies.

2. **2D flow field interference patterns and the proposal of the equivalent phase model.** The 2D parametric study reveals that the tandem arrangement has negligible impact on the anterior fin thrust, whereas the posterior fin thrust exhibits a pronounced sinusoidal periodic fluctuation with respect to both the fin spacing (Δx) and the phase difference ($\Delta\phi$). Based on flow field phenomena and dimensional analysis, this paper innovatively proposes a semi-empirical analytical formula centered on the "equivalent phase" as the core variable. This

model physically normalizes the hysteresis effect of the spatial convection of wake vortices and the temporal misalignment of mechanical control. Extensive validation by varying parameters such as phase difference, fin length, undulation amplitude, and frequency demonstrates that the model captures the evolution pattern of the posterior fin thrust with extremely high accuracy ($R^2 > 0.97$).

3. **Validation of the semi-empirical model's applicability in 3D flow fields.** To examine the cross-dimensional universality of the proposed theoretical framework, the model was further independently validated in the spatial and temporal dimensions using a 3D undulating fin model. The results show that despite the presence of significant spanwise diffusion and 3D vortex ring dissipation in the 3D flow field, the mathematical structure of the semi-empirical formula maintains high self-consistency. The formula not only accurately predicts the sinusoidal interference evolution trend of the posterior fin thrust with respect to fin spacing and phase difference in the 3D environment but also precisely identifies the optimal space–time coupling extreme points capable of achieving "thrust exceeding that of a single isolated fin."

The equivalent phase interference model proposed in this study articulates the physical mechanisms of hydrodynamic coupling between the anterior and posterior fins in mathematical terms. It provides an efficient, universal, and physically explicit theoretical evaluation paradigm for the space–time coupling optimization and array configuration design of future multi-body cooperative propulsion systems, possessing significant engineering guidance value.

However, this study still has certain limitations that require further breakthroughs. First, the proposed formula remains a semi-empirical model; although its undetermined coefficients can reflect changes in the flow state, it currently lacks pure analytical mathematical support derived directly from the Navier–Stokes equations or classical vortex dynamics. Second, to clearly isolate the interference mechanisms, this study only explored the purely undulatory propulsion mode. In contrast, real biomimetic vehicles typically employ composite motion modes coupling undulation and rolling. Introducing spanwise heaving will make the 3D wake interference mechanisms considerably more complex. Finally, due to current limitations in computational power, the calculations in this study primarily rely on simplified two-dimensional models. To overcome this bottleneck for three-dimensional undulating fins in future work, we plan to incorporate the wall-modeled large-eddy simulation (WMLES) framework developed by Cheng and coworkers Cheng et al. (2015); Gao et al. (2019); Cheng et al. (2022) into our computational code. Implementing this virtual wall model will significantly reduce the stringent near-wall grid resolution requirements, thereby making the high-fidelity calculation of fully 3D models much more computationally efficient and feasible.

Looking forward, exploring the universality of the "equivalent phase" framework represents a compelling frontier. While its applicability has been strictly validated for purely undulatory propulsion in 2D and 3D regimes, the underlying decoupling mechanism may transcend this specific motion. We hypothesize that the equivalent phase theory—unifying spatial convection hysteresis and temporal phase misalignment—could serve as a universal paradigm for a broader class of tandem biomimetic propulsion modes, such as pitching and heaving. Ultimately, this theoretical framework may prove widely applicable to quantifying wake interference and complex fluid interactions across diverse multi-body systems, laying a robust foundation for future theoretical and engineering endeavors.

CRediT authorship contribution statement

Wenzhuo Li: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Writing–original draft, Writing–review & editing. **Ming Xu:** Data curation, Investigation, Validation, review. **Shunxiang Pang:** Data curation, Investigation, Validation, review. **Fenghua Qin:** Conceptualization, Funding acquisition, Investigation, Project administration, Resources, Writing–review & editing. **Jun Jing:** Conceptualization, Data curation, Investigation, Validation, review. **Wan Cheng:** Data curation, Investigation, Validation, review. **Xisheng Luo:** Data curation, Investigation, Validation, review.

References

Arreola, V., Westneat, M., 1996. Mechanics of propulsion by multiple fins: kinematics of aquatic locomotion in the burrfish (chilomycterus schoepfi). Proceedings of the Royal Society of London. Series B: Biological Sciences 263, 1689–1696.

Bandyopadhyay, P.R., 2005. Trends in biorobotic autonomous undersea vehicles. IEEE Journal of Oceanic Engineering 30, 109–139.

Beal, D., Hover, F., Triantafyllou, M., Liao, J., Lauder, G., 2006. Passive propulsion in vortex wakes. Journal of fluid Mechanics 549, 385–402.

Becker, A., Masoud, H., Newbolt, J., Shelley, M., Ristroph, L., 2015. Hydrodynamic schooling of flapping swimmers. Nature communications 6, 8514.

Blake, R., 1983. Swimming in the electric eels and knifefishes. Canadian Journal of Zoology 61, 1432–1441.

Chan, W., Kang, T., Lee, Y., 2007. Experiments and identification of an ostraciiform fish robot, in: 2007 IEEE International Conference on Robotics and Biomimetics (ROBIO), IEEE. pp. 530–534.
Cheng, W., Pullin, D.I., Samtaney, R., 2015. Large-eddy simulation of separation and reattachment of a flat plate turbulent boundary layer. Journal of Fluid Mechanics 785, 78–108.
Cheng, W., Pullin, D.I., Samtaney, R., 2022. Wall-resolved and wall-modelled large-eddy simulation of plane couette flow. Journal of Fluid Mechanics 934, A19.
Costa, D., Callegari, M., Palmieri, G., Scaradozzi, D., Brocchini, M., Zitti, G., 2018a. Experimental setup for the validation of the bio-inspired thruster of an ostraciiform swimming robot, in: 2018 14th IEEE/ASME International Conference on Mechatronic and Embedded Systems and Applications (MESA), IEEE. pp. 1–6.
Costa, D., Franciolini, M., Palmieri, G., Crivellini, A., Scaradozzi, D., 2017. Computational fluid dynamics analysis and design of an ostraciiform swimming robot, in: 2017 IEEE International Conference on Robotics and Biomimetics (ROBIO), IEEE. pp. 135–140.
Costa, D., Palmieri, G., Palpacelli, M.C., Panebianco, L., Scaradozzi, D., 2018b. Design of a bio-inspired autonomous underwater robot, in: Journal of Intelligent & Robotic Systems, Springer. pp. 181–192.
Costa, D., Palmieri, G., Scaradozzi, D., Callegari, M., 2018c. Multi-body analysis of a bio-inspired underwater robot, in: The International Conference of IFToMM ITALY. Springer, pp. 240–248.
Dai, L., He, G., Zhang, X., Zhang, X., 2018. Stable formations of self-propelled fish-like swimmers induced by hydrodynamic interactions. Journal of The Royal Society Interface 15, 20180490.
Epstein, M., Colgate, J., MacIver, M., 2006. Generating thrust with a biologically-inspired robotic ribbon fin, in: 2006 IEEE/RSJ International Conference on Intelligent Robots and Systems, IEEE. pp. 2412–2417.
Gao, W., Zhang, W., Cheng, W., Samtaney, R., 2019. Wall-modelled large-eddy simulation of turbulent flow past airfoils. Journal of Fluid Mechanics 873, 174–210.
Guo, S., Fukuda, T., Asaka, K., 2003. A new type of fish-like underwater microrobot. IEEE/ASME Transactions on mechatronics 8, 136–141.
Hove, J., O'Bryan, L., Gordon, M., Webb, P., Weihs, D., 2001. Boxfishes (teleostei: Ostraciidae) as a model system for fishes swimming with many fins: kinematics. Journal of Experimental Biology 204, 1459–1471.
Hu, T., Shen, L., Lin, L., Xu, H., 2009. Biological inspirations, kinematics modeling, mechanism design and experiments on an undulating robotic fin inspired by gymnarchus niloticus. Mechanism and Machine Theory 44, 633–645.
Jia, L., Yin, X., 2008. Passive oscillations of two tandem flexible filaments in a flowing soap film. Physical review letters 100, 228104.
Kim, H., Lee, J., Chu, W., Ahn, S., 2017. Design and fabrication of soft morphing ray propulsor: undulator and oscillator. Soft robotics 4, 49–60.
Krishnadas, A., Ravichandran, S., Rajagopal, P., 2018. Analysis of biomimetic caudal fin shapes for optimal propulsive efficiency. Ocean Engineering 153, 132–142.
Kruusmaa, M., Fiorini, P., Megill, W., Vittorio, D., et al., 2014. Filose for svenning: a flow sensing bioinspired robot. IEEE Robotics & Automation Magazine 21, 51–62.
Latt, J., Malaspinas, O., Kontaxakis, D., Parmigiani, A., Lagrava, D., Brogi, F., Belgacem, M.B., Thorimbert, Y., Leclaire, S., Li, S., Marson, F., Lemus, J., Kotsalos, C., Conradin, R., Coreixas, C., Petkantchin, R., Raynaud, F., Beny, J., Chopard, B., 2021. Palabos: Parallel lattice boltzmann solver. Computers Mathematics with Applications 81, 334–350.
Lauder, G., Anderson, E., Tangorra, J., Madden, P., 2007. Fish biorobotics: kinematics and hydrodynamics of self-propulsion. Journal of Experimental Biology 210, 2767–2780.
Li, G., Ma, P., Fang, X., Li, G., Liu, G., Cheng, H., 2024. Study on thrust increase characteristics of separated undulating fins. Ocean Engineering 313, 119292.
Liao, J., Beal, D., Lauder, G., Triantafyllou, M., 2003. The kármán gait: novel body kinematics of rainbow trout swimming in a vortex street. Journal of Experimental Biology 206, 1059–1073.
Lighthill, J., 1990a. Biofluiddynamics of balistiform and gymnotiform locomotion. part 2. the pressure distribution arising in two-dimensional irrotational flow from a general symmetrical motion of a flexible flat plate normal to itself. Journal of Fluid Mechanics 213, 1–10.
Lighthill, J., 1990b. Biofluiddynamics of balistiform and gymnotiform locomotion. part 3. momentum enhancement in the presence of a body of elliptic cross-section. Journal of Fluid Mechanics 213, 11–20.
Lighthill, J., 1990c. Biofluiddynamics of balistiform and gymnotiform locomotion. part 4. short-wavelength limitations on momentum enhancement. Journal of Fluid Mechanics 213, 21–28.
Lighthill, J., Blake, R., 1990. Biofluiddynamics of balistiform and gymnotiform locomotion. part 1. biological background, and analysis by elongated-body theory. Journal of Fluid Mechanics 212, 183–207.
Liu, H., Curet, O., 2018. Swimming performance of a bio-inspired robotic vessel with undulating fin propulsion. Bioinspiration & biomimetics 13, 056006.
Low, K., 2006. Locomotion and depth control of robotic fish with modular undulating fins. International Journal of Automation and Computing 3, 348–357.
MacIver, M.A., Fontaine, E., Burdick, J.W., 2004. Designing future underwater vehicles: principles and mechanisms of the weakly electric fish. IEEE Journal of Oceanic Engineering 29, 651–659.
Mwaffo, V., Zhang, P., Cruz, S., Porfiri, M., 2017. Zebrafish swimming in the flow: a particle image velocimetry study. PeerJ 5, e4041.
Neveln, I., Bale, R., Bhalla, A., Curet, O., Patankar, N., MacIver, M., 2014. Undulating fins produce off-axis thrust and flow structures. Journal of Experimental Biology 217, 201–213.
Pan, D.y., Liu, H., Shao, X.m., 2010. Studies on the oscillation behavior of a flexible plate in the wake of a d-cylinder. Journal of Hydrodynamics 22, 132–137.
Pang, S., Qin, F., Shang, W., Cong, S., Zhang, S., 2021. Optimized design and investigation about propulsion of bionic tandem undulating fins i: Effect of phase difference. Ocean Engineering 239, 109842.

Ramananarivo, S., Fang, F., Oza, A., Zhang, J., Ristroph, L., 2016. Flow interactions lead to orderly formations of flapping wings in forward flight. Physical Review Fluids 1, 071201.

Salazar, R., Fuentes, V., Abdelkefi, A., 2018. Classification of biological and bioinspired aquatic systems: A review. Ocean Engineering 148, 75–114.

Sfakiotakis, M., Fasoulas, J., Kavoussanos, M., Arapis, M., 2015. Experimental investigation and propulsion control for a bio-inspired robotic undulatory fin. Robotica 33, 1062–1084.

Sfakiotakis, M., Lane, D.M., Davies, J., 1999. Review of fish swimming modes for aquatic locomotion. IEEE Journal of oceanic engineering 24, 237–252.

Shang, L., Wang, S., Tan, M., Cheng, L., 2012. Swimming locomotion modeling for biomimetic underwater vehicle with two undulating long-fins. Robotica 30, 913–923.

Shirgaonkar, A., Curet, O., Patankar, N., MacIver, M., 2008. The hydrodynamics of ribbon-fin propulsion during impulsive motion. Journal of Experimental Biology 211, 3490–3503.

Tian, F.B., Wang, W., Wu, J., Sui, Y., 2016. Swimming performance and vorticity structures of a mother–calf pair of fish. Computers & Fluids 124, 1–11.

Wang, R., Wang, S., Wang, Y., Tan, M., Yu, J., 2017. A paradigm for path following control of a ribbon-fin propelled biomimetic underwater vehicle. IEEE Transactions on Systems, Man, and Cybernetics: Systems 49, 482–493.

Wei, C., Li, S., Hu, Q., 2024. Hydrodynamic performance analysis of formations of dual three-dimensional undulating fins. Ocean Engineering 305, 117939.

Yuh, J., 2000. Design and control of autonomous underwater robots: A survey. Autonomous robots 8, 7–24.

Zhang, J., Bai, Y., Zhai, S., Gao, D., 2019. Numerical study on vortex structure of undulating fins in stationary water. Ocean Engineering 187, 106166.

Zhou, H., Hu, T., Xie, H., Zhang, D., Shen, L., 2010. Computational hydrodynamics and statistical modeling on biologically inspired undulating robotic fins: a two-dimensional study. Journal of Bionic Engineering 7, 66–76.

Zhu, X., He, G., Zhang, X., 2014. Flow-mediated interactions between two self-propelled flapping filaments in tandem configuration. Physical review letters 113, 238105.